

Question 1

Partially correct

Mark 6.40 out of 8.00

Which statements about Unsupervised Machine Learning are true?

Multiple answers are possible!

Select one or more:

- ☒ a. k-nearest neighbors is a nonparametric model. ✓
- ☒ b. In parametric models, every parameter (vector) realization represents a model. ✓
- ☒ c. Examples of unsupervised methods are: PCA, ICA, Factor Analysis, k-means clustering, and mixture models. ✓
- ☐ d. Generative models do not aim at modeling the data creation process.
- ☐ e. The model selection procedure is called "training" or "learning".
- ☒ f. Unsupervised Machine Learning algorithms learn patterns from unlabeled data. ✓
- ☐ g. Underfitting means that a model is adapted to special training characteristics such as noisy measurements or outliers.
- ☐ h. Projection methods project data into a higher-dimensional space while keeping their neighborhoods.

Die Antwort ist teilweise richtig.

You have correctly selected 4.

The correct answers are: Unsupervised Machine Learning algorithms learn patterns from unlabeled data., Examples of unsupervised methods are: PCA, ICA, Factor Analysis, k-means clustering, and mixture models., In parametric models, every parameter (vector) realization represents a model., k-nearest neighbors is a nonparametric model., The model selection procedure is called "training" or "learning".

Partially correct

Mark 6.00 out of 8.00

Which statements about Expectation Maximization (EM) and Maximum Entropy are correct?

Multiple answers are possible!

Select one or more:

- ☒ a. In the M-step, the hidden variables are updated. ✖
- ☒ b. Maximum Entropy requires minimal prior assumptions to assign a-priori-probabilities. ✔
- ☒ c. The E-step increases the lower bound on the log-likelihood. ✔
- ☒ d. The EM algorithm can be applied to hidden Markov models and mixture models. ✔
- ☐ e. The entropy of a probability distribution is always larger equal 0 and smaller equal 1.
- ☐ f. The E-step increases the upper bound on the log-likelihood.
- ☐ g. In the E-step, the model parameters are updated.
- ☒ h. In EM, one deals with problems that have hidden variables as well as model parameters. ✔

Die Antwort ist teilweise richtig.

You have selected too many options.

The correct answers are: In EM, one deals with problems that have hidden variables as well as model parameters., The EM algorithm can be applied to hidden Markov models and mixture models., The E-step increases the lower bound on the log-likelihood., Maximum Entropy requires minimal prior assumptions to assign a-priori-probabilities.

Which statements about Principal Component Analysis (PCA) are correct?

Multiple answers are possible!

Select one or more:

- ☐ a. PCA is a data projection method that aims at minimizing the variance in the projected data on a lower dimensional space.
- ☐ b. One can use the singular value decomposition of the data matrix  $\mathbf{X}$  to perform PCA.
- ☐ c. The projected data is given by  $\mathbf{Y} = \mathbf{X}\mathbf{U}$ , where  $\mathbf{X}$  is the (centered) original data and  $\mathbf{U}$  is the matrix of eigenvectors.
- ☒ d. The PCA solution is unique up to signs, given that eigenvalues correspond to unique eigenvectors. ✓
- ☐ e. The sample mean of the  $k$ -th projection is equal to the  $k$ -th eigenvalue of the sample covariance matrix.
- ☐ f. The matrix of eigenvectors  $\mathbf{U}$  is symmetric and positive-semidefinite.
- ☐ g. The eigenvector corresponding to the smallest eigenvalue of the eigendecomposition of the sample covariance matrix is called the first principal component.
- ☐ h. Common applications of PCA include data compression, data visualization and data pre-processing.

Die Antwort ist teilweise richtig.

You have correctly selected 1.

The correct answers are: Common applications of PCA include data compression, data visualization and data pre-processing. ,

The projected data is given by  $\mathbf{Y} = \mathbf{X}\mathbf{U}$ , where  $\mathbf{X}$  is the (centered) original data and  $\mathbf{U}$  is the matrix of eigenvectors.

, One can use the singular value decomposition of the data matrix  $\mathbf{X}$  to perform PCA.

, The PCA solution is unique up to signs, given that eigenvalues correspond to unique eigenvectors.

**Question 4**

Incorrect

Mark 0.00 out of 9.00

You are given a dataset with  $n$  samples and three features. The covariance matrix  $\mathbf{C}$  has eigenvalues  $(\lambda_1, \lambda_2, \lambda_3) = (5, 3, 1)$  and associated normalized eigenvectors  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ ,  $\|\mathbf{v}_i\| = 1$ . Calculate

$$A = (2\mathbf{v}_1^T + \mathbf{v}_2^T + 3\mathbf{v}_3^T)\mathbf{C}(3\mathbf{v}_2 + \mathbf{v}_3)$$

A=



Tick the correct statements about Kernel PCA.

We stick to the notation from the lecture, i.e. the data points are denoted by  $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathcal{X} = \mathbb{R}^m$ , the kernel by  $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ , and the feature map by  $\Phi : \mathcal{X} \rightarrow \mathcal{H}$ . The feature covariance matrix is defined by  $\mathbf{C} = \frac{1}{n} \sum_{i=1}^n \Phi(\mathbf{x}_i) \Phi(\mathbf{x}_i)^T$  and the Gram matrix  $\mathbf{K}$  by  $K_{ij} = \Phi(\mathbf{x}_i)^T \Phi(\mathbf{x}_j)$  for  $i, j = 1, \dots, n$ .

Select one or more:

- ☒ a. The main task of kernel PCA is to solve the eigenvalue problem of  $\mathbf{C}$  in feature space. This can be achieved by solving the eigenvalue problem of  $\mathbf{K}$  in a Euclidean space. ✓
- ☐ b. For kernel PCA, it is essential to center  $\mathbf{K}$  before computing its eigendecomposition, since the eigenvalues of  $\mathbf{K}$  can be in general be negative, which leads to inconclusive results.
- ☒ c. In general,  $\Phi$  is a nonlinear map that maps into a higher dimensional feature space  $\mathcal{H}$ . ✓
- ☐ d.  $\mathbf{K}$  and  $\mathbf{C}$  always have the same dimensionality and the same rank.
- ☒ e. Linear PCA can be seen as a special case of kernel PCA by using  $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \mathbf{y}$  as the associated kernel function. ✓
- ☒ f. To compute the projection of a data point  $\mathbf{x}$  onto the eigenvectors of  $\mathbf{K}$ , you need to center  $\mathbf{x}$  and  $\mathbf{K}$ , as well as to normalize the eigenvectors of  $\mathbf{K}$ . ✓
- ☐ g. A big disadvantage of Kernel PCA is that  $\Phi$  always needs to be computed explicitly, which is very costly.

Die Antwort ist richtig.

The correct answers are: In general,  $\Phi$  is a nonlinear map that maps into a higher dimensional feature space  $\mathcal{H}$ .

, To compute the projection of a data point  $\mathbf{x}$  onto the eigenvectors of  $\mathbf{K}$ , you need to center  $\mathbf{x}$  and  $\mathbf{K}$ , as well as to normalize the eigenvectors of  $\mathbf{K}$ .

, Linear PCA can be seen as a special case of kernel PCA by using  $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \mathbf{y}$  as the associated kernel function.

, The main task of kernel PCA is to solve the eigenvalue problem of  $\mathbf{C}$  in feature space. This can be achieved by solving the eigenvalue problem of  $\mathbf{K}$  in a Euclidean space.

Question 6

Partially correct

Mark 3.73 out of 8.00

Which statements about Independent Component Analysis (ICA) are correct?

Multiple answers are possible!

Select one or more:

- ☐ a. The ICA solution is unique.
- ☒ b. In general, ICA components are neither ranked nor orthogonal. ✓
- ☐ c. For ICA, one needs more sources than observations.
- ☐ d. Non-Gaussianity can be measured by the second cumulant.
- ☐ e. Criteria for independence are mutual information between components and non-Gaussianity of components.
- ☒ f. Fast ICA relies on whitening and rotating the data. ✓
- ☒ g. Maximizing the negentropy maximizes the mutual information. ✗
- ☐ h. Infomax maximizes the mutual information between components.

Die Antwort ist teilweise richtig.

You have correctly selected 2.

The correct answers are: In general, ICA components are neither ranked nor orthogonal., Criteria for independence are mutual information between components and non-Gaussianity of components., Fast ICA relies on whitening and rotating the data.

**Question 7**

Incorrect

Mark 0.00 out of 8.00

In factor analysis part of the the covariance function is formed by

$$\mathbf{C_U} = \mathbf{U}\mathbf{U}^T = \sum_{i=1}^l \mathbf{u}_i \mathbf{u}_i^T,$$

where  $\mathbf{U} \in \mathbb{R}^{m \times l}$  and  $\mathbf{u}_i$  is the  $i$ -th column of  $\mathbf{U}$  and  $l < m$

Which constraints does this impose on the  $\mathbf{C_U}$ ?

- ☒ a. The maximum rank of the matrix is  $l$ . ✓
- ☒ b. The number of zero eigenvalues of  $\mathbf{C_U}$  is greater or equal to  $m - l$ . ✓
- ☒ c. Any covariance matrix in  $m$  dimensions can be written in the form above. ✗
- ☐ d.  $\mathbf{C_U}$  is singular.

Your answer is incorrect.

The correct answers are:

The maximum rank of the matrix is  $l$ .

,

The number of zero eigenvalues of  $\mathbf{C_U}$  is greater or equal to  $m - l$ .

,

$\mathbf{C_U}$  is singular

**Question 8**

Correct

Mark 8.00 out of 8.00

Assume you have two random variables  $Y$  and  $Z$ , which are uniformly distributed on  $[0, \theta_Y]$  and  $[0, \theta_Z]$ , respectively, i.e.

$$p_Y(x) = \begin{cases} \frac{1}{\theta_Y} & \text{if } x \in [0, \theta_Y] \\ 0 & \text{else.} \end{cases}$$

$$p_Z(x) = \begin{cases} \frac{1}{\theta_Z} & \text{if } x \in [0, \theta_Z] \\ 0 & \text{else.} \end{cases}$$

For  $\theta_Y = e^5$  and  $\theta_Z = e^{10}$ , compute the KL divergence

$$D_{\text{KL}}(p_Y || p_Z) = \int_{-\infty}^{\infty} p_Y(x) \log[p_Y(x)/p_Z(x)] \, dx.$$

As for "log", use the natural logarithm.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

$$D_{\text{KL}}(p_Y || p_Z) =$$





Question 9

Incorrect

Mark 0.00 out of 8.00

Which statements about t-distributed Stochastic Neighborhood Embedding are correct? t-SNE...

- ☒ a. ...models the neighborhood in both observation and embedding space by a Gaussian distribution. ✖
- ☒ b. ...minimizes the Kullback-Leibler divergence between distributions in observation space and embedding space. ✔
- ☒ c. ...models the neighborhood of a data point in the observation space proportional to a Student's t-distribution. ✖
- ☒ d. ...is usually trained by gradient descent. ✔

Your answer is incorrect.

The correct answers are:

...minimizes the Kullback-Leibler divergence between distributions in observation space and embedding space.,

...is usually trained by gradient descent.

Assume you want to do  $k$ -means on the following 2-dimensional dataset:

$$X = \{(1, 0), (0, 1), (-1, 0), (0, -1), (2, 0), (2, 3), (5, 0)\}.$$

The initial cluster centers are  $\mu = (0, 1)$  and  $\nu = (2, 2)$ .

Compute the coordinates of the new cluster centers  $\mu^{\text{new}} = (\mu_1^{\text{new}}, \mu_2^{\text{new}})$  and  $\nu^{\text{new}} = (\nu_1^{\text{new}}, \nu_2^{\text{new}})$  after one iteration of  $k$ -means.

Hint: Make a sketch first. The problem can mainly be solved graphically.

$$\mu_1^{\text{new}} =$$

✓ ,  $\mu_2^{\text{new}} =$

✓

$$\nu_1^{\text{new}} =$$

✓ ,  $\nu_2^{\text{new}} =$

✓

Mark 0.00 out of 10.00

Mixturetown has two bakers call Albert (A) and Bibi (B). Both bake bread where the weights of the loafs they produce are distributed according to two Gaussian distributions.

The weight distributions of Albert's and Bibi's loafs are  $f(x; \mu_A, \sigma_A)$  and  $f(x; \mu_B, \sigma_B)$  respectively, where  $f(x; \mu, \sigma)$  is the probability density function of a normal distribution with mean  $\mu$  and standard deviation  $\sigma$ .

The means and standard deviations are known to be  $\mu_A = 1$  kg,  $\sigma_A = 0.1$  kg,  $\mu_B = 1.5$  kg,  $\sigma_B = 0.2$  kg.

Thus the distribution of loaf weights in Mixturetown is well described by a Mixture of two Gaussians:

$$p(x) = \alpha_A f(x; \mu_A, \sigma_A) + \alpha_B f(x; \mu_B, \sigma_B)$$

75 per cent of the loafs in Mixturetown are made by Bibi.

Somebody is so kind to give you a loaf of bread that weighs 1.1 kg.

Determine the probability that Bibi made this loaf of bread (up to two decimal places).

Some relevant quantities are  $f(1.1; 1, 0.1) = 2.41971$ ,  $f(1.1; 1.5, 0.2) = 0.26995$ .

Answer:  ✖

The correct answer is: 0.25076529093399